

1.(2 pts) If A is a 3 x 5 matrix and B is a 5 x 2 matrix, what is the size of the matrix product AB?

2. (2 pts) If A is a 3 x 5 matrix and the product AB is a 3 x 5 matrix, what must be the size of the matrix B?

3. (4 pts) If $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ compute the matrix products shown below.

a) $\mathbf{u}^T \mathbf{v} =$

b) $\mathbf{u} \mathbf{v}^T$

4. (4 pts) Complete the following sentence.

a) What is the determinant of the matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$?

b) How is the determinant of A used to find A^{-1} ?

5. (3 pts) What can you say about the solution to the matrix equation $A\mathbf{x} = \mathbf{b}$ if A is an invertible matrix?